Quang Truong

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Research Interests

Graph Representation Learning, Topological Deep Learning, Machine Learning

Education

Michigan State University — *East Lansing, MI*
Ph.D. in Computer Science — *Aug. 2024 - Present*

Dartmouth College — *Hanover, NH*
M.S. in Engineering Sciences — *Sep. 2022 - Jun. 2024*

Research Experience

pyt-team — *Remote*
Contributor — *Aug. 2023 - Oct. 2023*

- Initiate the development of Path Complex package for the TopoNetX library.

LISP Lab, Dartmouth College — *Hanover, NH*
*Graduate Student (Advisor: **Peter Chin**)* — *Sep. 2022 - Jun. 2024*

- Pioneer a novel branch of TDL that relies on simple paths, which significantly enhance generalizability of Topological Graph Neural Networks.
- Prove that the Path WL test inherits and generalizes theoretical properties of Simplicial WL and Cell WL tests.
- Develop Path Isomorphism Network, a state-of-the-art TDL model, which achieves remarkable results on PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).

Vision Lab, University of Illinois at Chicago — *Remote*
*Research Intern (Advisor: **Wei Tang**)* — *May 2021 - Feb. 2022*

- Innovated a context-aware 3D object detection architecture, underpinned by a self-attention mechanism via non-local blocks, that captures object-wise and object-context relationships.
- Enhanced model performance by incorporating contrastive learning techniques to align allocentric, egocentric, and semantic feature embeddings.

ML Lab, Texas Christian University — *Fort Worth, TX*
*Undergraduate Research Assistant (Advisor: **Bo Mei**)* — *May 2019 - May 2021*

- Architected a Vehicle Re-identification pipeline empowered by Generative Adversarial Networks (GANs) for adaptive domain learning.
- Derived a novel filtering algorithm that substantially reduces the number of training samples without sacrificing performance.

Selected Publications

‡ indicates equal contribution.

[1] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains
Mustafa Hajij‡, Mathilde Papillon‡, Florian Frantzen‡, ... **Quang Truong**, and ... Nina Miolane. **2024a**
Under review at the Journal of Machine Learning Research (JMLR). arXiv: 2402.02441 [cs.LG].

[2] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes
Quang Truong and Peter Chin. **2024b**
AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence. DOI: 10.1609/aaai.v38i14.29463.

[3] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification
Quang Truong and Bo Mei. **2021**
ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems. DOI: 10.1109/ITSC48978.2021.9564945.

[4] Image-based Vehicle Re-identification Model with Adaptive Attention Modules and Metadata Re-ranking
Quang Truong, Hy Dang, Zhankai Ye, Minh Nguyen, and Bo Mei. **2020**
The Boller Review: A TCU Journal of Undergraduate Research and Creativity (Vol. 5). TCU Press. DOI: 10.18776/tcu/br/5/130.

Presentations

Invited Presentations

- Topologically-driven Neural Networks, *HPCC Special Seminar: Decoding Computational Challenges, the Advanced Institute of Interdisciplinary Science and Technology - HCMUT, VNU-HCM*. **2024**
- Domain-invariant Network for Vehicle Re-identification, *Annual Industrial Board Meeting of TCU Department of Computer Science*. **2020**

Conference Presentations

- Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes, *AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*. **2024**
- Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-identification, *ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*. **2021**

Teaching Experience

Dartmouth College

- Teaching Assistant, *ENGS 96: Mathematics for Machine Learning*. **Winter 2024**

Honors

Grants

- Research Grant for Domain-invariant Network for Vehicle Re-identification ($1471), *TCU SERC Undergraduate Research Grant*. **2020**
- Research Grant for AI-2-Go ($1500), *TCU SERC Undergraduate Research Grant*. **2019**
- Research Grant for Housing Price Prediction ($1500), *TCU SERC Undergraduate Research Grant*. **2019**

Awards

- VEF Fellow, *VEF 2.0 Program, VEF Fellows and Scholars Association*. **2021**
- Vingroup Scholarship Recipient, *Master's & Ph.D. Scholarship Program, VinUniversity*. **2021**
- Best Undergraduate Research Poster, *TCU Student Research Symposium [URL]*. **2019, 2021**
- TCU Scholar, *Texas Christian University*. **2020, 2021**
- Dean's List, *Texas Christian University*. **2019-2021**
- Bronze Medal, *ACM-ICPC South Central USA Regional Contest [URL]*. **2019**
- 1st Prize, *Calculus Bee [URL]*. **2019**
- Transfer Faculty Scholarship ($196,000 for four years), *Texas Christian University*. **2019**
- President's Scholar, *Mississippi State University*. **2017, 2018**
- Freshmen Academic Excellence Scholarship ($54,000 for four years), *Mississippi State University*. **2017**

Skills

Programming	Python, C/C++, Java, SQL, R, Scala
Machine Learning	Pytorch, Pytorch Geometric, NetworkX, TopoNetX, Pytorch Lightning, Hydra, WandB, Scikit-learn, Tensorflow, Keras, Matplotlib, Numpy, Pandas

Relevant Coursework

Engineering	Signal Processing, Fourier Transforms and Complex Variables, Statistical Methods in Engineering
Computer Science	Multi-modalities Generative AI, Network Science and Complex Systems, Deep Learning, Data Mining and Visualization, Analysis of Algorithm, Operating Systems, Object-oriented Programming, Database Systems, Programming Language Concepts
Mathematics	Mathematics for Machine Learning, Game Theory, Applied Linear Algebra, Probability, Statistics, Discrete Mathematics, Calculus